Lecture 4: Numerically Solve RBC Model

Chengyuan He

March 2026

Solve LOM

$$\hat{k}_t = \frac{1}{\beta}\hat{k}_{t-1} + \frac{\tilde{\delta}}{\alpha\beta}\hat{z}_t - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right)\hat{c}_t$$

$$\sigma E_t \hat{c}_{t+1} + \tilde{\delta}(1-\alpha)\hat{k}_t - \tilde{\delta} E_t \hat{z}_{t+1} = \sigma \hat{c}_t$$

$$\hat{z}_{t+1} = \psi \hat{z}_t + \varepsilon_t$$

State variables: $\hat{k}_{t-1}$ and shock $\hat{z}_t$.

$\hat{k}_{t-1}$ is predetermined.

The dynamics of the model can be described as a recursive law of motion in terms of states.

LOM

$$\begin{cases} \hat{k}_t = V_{kk}\hat{k}_{t-1} + V_{kz}\hat{z}_t, \\ \hat{c}_t = V_{ck}\hat{k}_{t-1} + V_{cz}\hat{z}_t, \end{cases}$$

and solve for $(V_{kk}, V_{kz}, V_{ck}, V_{cz})$ by the method of undetermined coefficients.

1)

$$\hat{z}_{t+1} = \psi \hat{z}_t + \varepsilon_t \quad \Rightarrow \quad E_t \hat{z}_{t+1} = \psi \hat{z}_t.$$

Since $E_t(\varepsilon_t) = 0$, and $\varepsilon_t \sim iid\, N(0, \sigma^2)$,

$$E_t(\hat{z}_t) = \hat{z}_t,$$

because Z_t is already realized in period t, hence belongs to period-t's information set.

2) From

$$\hat{k}_t = \frac{1}{\beta}\hat{k}_{t-1} + \frac{\tilde{\delta}}{\alpha\beta}\hat{z}_t - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right)\hat{c}_t,$$

substitute

$$\hat{k}_t = V_{kk}\hat{k}_{t-1} + V_{kz}\hat{z}_t, \qquad \hat{c}_t = V_{ck}\hat{k}_{t-1} + V_{cz}\hat{z}_t,$$

to get

$$V_{kk}\hat{k}_{t-1} + V_{kz}\hat{z}_t = \frac{1}{\beta}\hat{k}_{t-1} - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right)\left(V_{ck}\hat{k}_{t-1} + V_{cz}\hat{z}_t\right) + \frac{\tilde{\delta}}{\alpha\beta}\hat{z}_t.$$

Collecting coefficients:

$$\left[\frac{1}{\beta} - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right) V_{ck} - V_{kk}\right] \hat{k}_{t-1} + \left[\frac{\tilde{\delta}}{\alpha\beta} - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right) V_{cz} - V_{kz}\right] \hat{z}_t = 0.$$

Hence

$$\boxed{V_{kk} = \frac{1}{\beta} - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right) V_{ck}} \tag{1}$$

and

$$\boxed{V_{kz} = \frac{\tilde{\delta}}{\alpha\beta} - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right) V_{cz}} \tag{2}$$

3) From

$$\sigma E_t \hat{c}_{t+1} + \tilde{\delta}(1-\alpha)\hat{k}_t - \tilde{\delta} E_t \hat{z}_{t+1} = \sigma \hat{c}_t,$$

substitute

$$\hat{c}_{t+1} = V_{ck}\hat{k}_t + V_{cz}\hat{z}_{t+1}, \qquad \hat{k}_t = V_{kk}\hat{k}_{t-1} + V_{kz}\hat{z}_t, \qquad E_t\hat{z}_{t+1} = \psi\hat{z}_t.$$

Then

$$\sigma E_t\left[V_{ck}\hat{k}_t + V_{cz}\hat{z}_{t+1}\right] + \tilde{\delta}(1-\alpha)\hat{k}_t - \tilde{\delta}\psi\hat{z}_t = \sigma\left(V_{ck}\hat{k}_{t-1} + V_{cz}\hat{z}_t\right).$$

That is,

$$\sigma E_t\left[V_{ck}\left(V_{kk}\hat{k}_{t-1} + V_{kz}\hat{z}_t\right) + V_{cz}\hat{z}_{t+1}\right] + \tilde{\delta}(1-\alpha)\left(V_{kk}\hat{k}_{t-1} + V_{kz}\hat{z}_t\right) - \tilde{\delta}\psi\hat{z}_t = \sigma V_{ck}\hat{k}_{t-1} + \sigma V_{cz}\hat{z}_t.$$

Collecting coefficients:

$$\left[\sigma V_{ck}V_{kk} + \tilde{\delta}(1-\alpha)V_{kk} - \sigma V_{ck}\right]\hat{k}_{t-1} + \left[\sigma(V_{ck}V_{kz} + V_{cz}\psi) + \tilde{\delta}(1-\alpha)V_{kz} - \tilde{\delta}\psi - \sigma V_{cz}\right]\hat{z}_t = 0.$$

Hence

$$\boxed{\sigma V_{ck}(1 - V_{kk}) = \tilde{\delta}(1-\alpha)V_{kk}} \tag{3}$$

and

$$\boxed{\sigma V_{cz}(1-\psi) = \left[\sigma V_{ck} + \tilde{\delta}(1-\alpha)\right]V_{kz} - \tilde{\delta}\psi} \tag{4}$$

Solve the 4 Equations for 4 Unknowns

Collect the four equations and solve for the four unknowns.

(i) (1) and (3) are two equations in (V_{kk}, V_{ck}). Eliminate V_{ck}.

From (1),

$$V_{ck} = \frac{\frac{1}{\beta} - V_{kk}}{\frac{\tilde{\delta}}{\alpha\beta} - \delta}.$$

Substitute into (3):

$$\sigma \left(\frac{\frac{1}{\beta} - V_{kk}}{\frac{\tilde{\delta}}{\alpha\beta} - \delta} \right) (1 - V_{kk}) = \tilde{\delta}(1-\alpha) V_{kk}.$$

So

$$\frac{\sigma}{\beta} - \frac{\sigma}{\beta} V_{kk} - \sigma V_{kk} + \sigma V_{kk}^2 = \tilde{\delta}(1-\alpha) \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta \right) V_{kk}.$$

Equivalently,

$$V_{kk}^2 - \left[1 + \frac{1}{\beta} + \frac{\tilde{\delta}(1-\alpha)}{\sigma} \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta \right) \right] V_{kk} + \frac{1}{\beta} = 0.$$

Let

$$\gamma = 1 + \frac{1}{\beta} + \frac{\tilde{\delta}(1-\alpha)}{\sigma} \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta \right).$$

Then the two roots are

$$V_{kk,1} = \frac{\gamma - \sqrt{\gamma^2 - \frac{4}{\beta}}}{2} = \frac{\gamma}{2} - \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}} > 0,$$

$$V_{kk,2} = \frac{\gamma + \sqrt{\gamma^2 - \frac{4}{\beta}}}{2} = \frac{\gamma}{2} + \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}} > 0.$$

We need to choose the stable root of the linear rational expectations (LRE) system.

Since

$$\gamma = 1 + \frac{1}{\beta} + \frac{\tilde{\delta}(1-\alpha)}{\sigma} \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta \right) > 1 + \frac{1}{\beta},$$

we have

$$(1 - V_{kk,1})(1 - V_{kk,2}) = 1 + V_{kk,1}V_{kk,2} - (V_{kk,1} + V_{kk,2}) = 1 + \frac{1}{\beta} - \gamma < 0.$$

Hence

$$0 < V_{kk,1} < 1 < V_{kk,2}.$$

So $V_{kk,2}$ is the explosive root and should be discarded.

To see this, consider the shutdown-shock case $\hat{z}_t = 0$:

$$\hat{k}_t = V_{kk} \hat{k}_{t-1}.$$

Then V_{kk} is the speed of convergence:

$$|V_{kk}| < 1 \;\Rightarrow\; \text{deviation in capital shrinks over time, economy returns to S.S.,}$$

$$|V_{kk}| > 1 \;\Rightarrow\; \text{deviation in capital grows over time, economy explodes and violates TVC.}$$

By the Blanchard–Kahn condition, there is one predetermined state $\hat{k}_{t-1}$ and one jump variable $\hat{c}_t$, so the number of stable roots must equal the number of predetermined states. Therefore, exactly one stable root is needed.

$$\boxed{\text{Drop } V_{kk,2};\ \text{only } V_{kk,1} \text{ is admissible.}}$$

Thus

$$V_{kk} = \frac{\gamma}{2} - \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}}.$$

And from (1),

$$V_{ck} = \frac{1}{\frac{\tilde{\delta}}{\alpha\beta} - \delta}\left(\frac{1}{\beta} - \frac{\gamma}{2} + \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}}\right).$$

(ii) (2) and (4) are two equations in (V_{kz}, V_{cz}). Solve for V_{cz} by substituting V_{kz} out.

Starting from

$$\sigma V_{cz}(1-\psi) = \left[\sigma V_{ck} + \tilde{\delta}(1-\alpha)\right]V_{kz} - \tilde{\delta}\psi,$$

and

$$V_{kz} = \frac{\tilde{\delta}}{\alpha\beta} - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right)V_{cz},$$

substitute the second into the first:

$$\sigma V_{cz}(1-\psi) = \left[\sigma\frac{1}{\frac{\tilde{\delta}}{\alpha\beta} - \delta}\left(\frac{1}{\beta} - \frac{\gamma}{2} + \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}}\right) + \tilde{\delta}(1-\alpha)\right]\left[\frac{\tilde{\delta}}{\alpha\beta} - \left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right)V_{cz}\right] - \tilde{\delta}\psi$$

$$\Rightarrow \quad V_{cz} = \frac{\sigma\frac{1}{\frac{\tilde{\delta}}{\alpha\beta}-\delta}\left(\frac{1}{\beta} - \frac{\gamma}{2} + \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}}\right) + \tilde{\delta}(1-\alpha) - \alpha\beta\psi}{\sigma(1-\psi) + \left[\sigma\frac{1}{\frac{\tilde{\delta}}{\alpha\beta}-\delta}\left(\frac{1}{\beta} - \frac{\gamma}{2} + \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}}\right) + \tilde{\delta}(1-\alpha)\right]\left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right)} \times \frac{\tilde{\delta}}{\alpha\beta}.$$

$$\Rightarrow \quad V_{kz} = \frac{\tilde{\delta}}{\alpha\beta} - \frac{\tilde{\delta}}{\alpha\beta}\left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right) \cdot \frac{\sigma\frac{1}{\frac{\tilde{\delta}}{\alpha\beta}-\delta}\left(\frac{1}{\beta} - \frac{\gamma}{2} + \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}}\right) + \tilde{\delta}(1-\alpha) - \alpha\beta\psi}{\sigma(1-\psi) + \left[\sigma\frac{1}{\frac{\tilde{\delta}}{\alpha\beta}-\delta}\left(\frac{1}{\beta} - \frac{\gamma}{2} + \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}}\right) + \tilde{\delta}(1-\alpha)\right]\left(\frac{\tilde{\delta}}{\alpha\beta} - \delta\right)}.$$

Now we have:

$$\begin{cases} \hat{k}_t = V_{kk}\hat{k}_{t-1} + V_{kz}\hat{z}_t, \\ \hat{c}_t = V_{ck}\hat{k}_{t-1} + V_{cz}\hat{z}_t, \\ \hat{z}_t = \psi\hat{z}_{t-1} + \varepsilon_t. \end{cases}$$

Where,

$$V_{kk} = \frac{\gamma}{2} - \sqrt{\left(\frac{\gamma}{2}\right)^2 - \frac{1}{\beta}}, \qquad V_{ck} = \frac{\bar{K}}{\bar{C}}\left(\frac{1}{\beta} - V_{kk}\right),$$

$$V_{cz} = \frac{\sigma V_{ck} + \tilde{\delta}(1-\alpha) - \alpha\beta\psi}{\sigma(1-\psi) + \left[\sigma V_{ck} + \tilde{\delta}(1-\alpha)\right]\frac{\bar{C}}{\bar{K}}} \times \frac{\tilde{\delta}}{\alpha\beta},$$

$$V_{kz} = \frac{\tilde{\delta}}{\alpha\beta} - \frac{\bar{C}}{\bar{K}}V_{cz}.$$

Feed in quarterly parameter values

$\beta = 0.99$	matches a 4.1% annual rate: $r_q = \frac{1}{\beta} - 1 \approx 1.01\%$, $\quad r_a \approx (1+r_q)^4 - 1 \approx 4.1\%$.
$\alpha = 0.36$	matches the capital income share in the data.
$\sigma = 1$	log utility. Usually $\sigma = 2$ implies IES $= \frac{1}{\sigma} = 0.5$.
$\delta = 0.025$	capital depreciates by 2.5% per quarter (about 10% annually).
$\bar{Z} = 1$	steady-state technology level is normalized.
$\psi = 0.95$	persistence of the productivity shock, i.e. how long ε_t lasts.

Then we can compute $(V_{kk}, V_{cz}, V_{kz}, V_{ck})$.

Impulse response function (IRF) Shock ε_t and trace how variables respond.

e.g. Trace out all variables for $\varepsilon_1 = 1$, $\varepsilon_t = 0$ for $t \geq 2$, start from S.S. (one-time temporary).

IRF and Half-life

Since

$$\hat{z}_t = \psi\hat{z}_{t-1} + \varepsilon_t,$$

we have

$$\hat{z}_1 = \psi\hat{z}_0 + \varepsilon_1 = \varepsilon_1,$$

since we start from steady state and $\hat{z}_0 = 0$. (Recall a hat means deviation from steady state.)

Also,

$$\hat{z}_2 = \psi\hat{z}_1 + \varepsilon_2 = \psi\hat{z}_1 = \psi\varepsilon_1,$$

and in general

$$\hat{z}_t = \psi^{t-1}\varepsilon_1.$$

When $|\psi| < 1$, the shock dies out as $t \to \infty$.

For capital,

$$\hat{k}_1 = V_{kk}\hat{k}_0 + V_{kz}\hat{z}_1 = V_{kz}\varepsilon_1,$$

since $\hat{k}_0 = 0$ at steady state.

Then

$$\hat{k}_2 = V_{kk}\hat{k}_1 + V_{kz}\hat{z}_2 = V_{kk}V_{kz}\varepsilon_1 + V_{kz}\psi\varepsilon_1 = (V_{kk} + \psi)V_{kz}\varepsilon_1.$$

In general,

$$\hat{k}_t = V_{kk}\hat{k}_{t-1} + V_{kz}\psi^{t-1}\varepsilon_1 = \sum_{i=0}^{t-1} V_{kk}^{i}\psi^{t-i-1}V_{kz}\varepsilon_1.$$

V_{kk}^i and V_{kz} are constant. Each past TFP effect ψ^{t-i-1} is stored/propagated through capital at rate V_{kk}^i, and V_{kz} translates the TFP shock into capital.

How to measure speed of convergence? Use half-life. We have

$$\hat{k}_{t+h} = V_{kk}^h \hat{k}_t.$$

The half-life is defined by

$$\frac{\hat{k}_{t+h_{1/2}}}{\hat{k}_t} = \frac{1}{2} \quad \Rightarrow \quad V_{kk}^{h_{1/2}} = \frac{1}{2} \quad \Rightarrow \quad \boxed{h_{1/2}^{(k)} = \frac{\ln(1/2)}{\ln(V_{kk})}} \qquad (0 < V_{kk} < 1).$$

For the TFP shock,

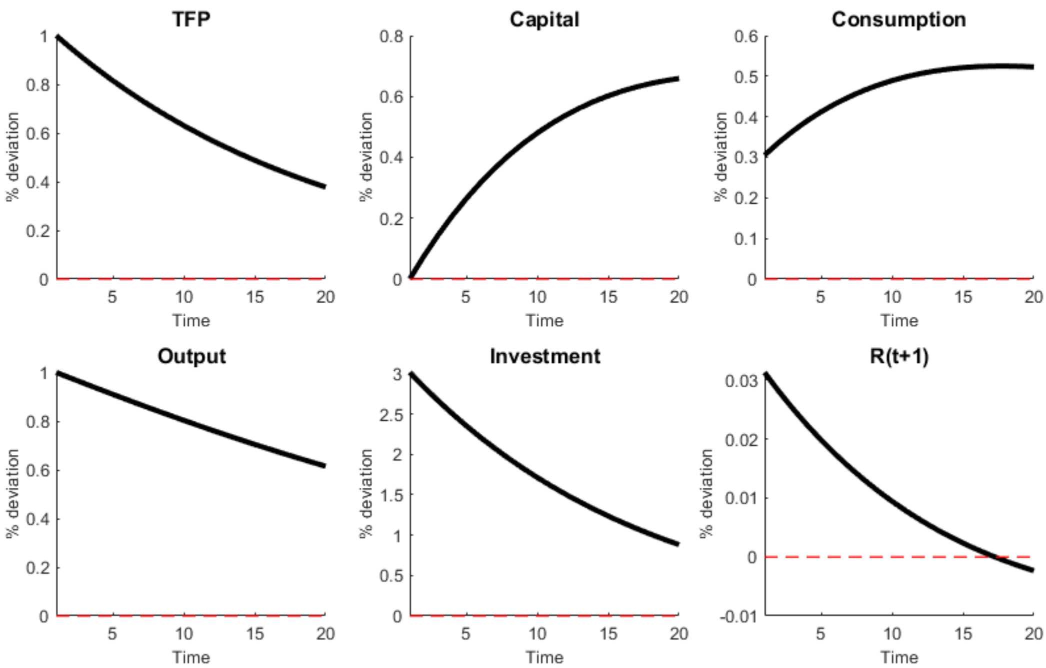
$$\boxed{h_{1/2}^{(z)} = \frac{\ln(1/2)}{\ln(\psi)}}.$$

Examples: if $V_{kk} = 0.95$,

$$h_{1/2}^{(k)} \approx 13.5 \text{ quarters} \approx 3.4 \text{ years}.$$

If $V_{kk} = 0.9$,

$$h_{1/2}^{(k)} \approx 6.6 \text{ quarters} \approx 1.6 \text{ years}.$$



Question: why is capital hump-shaped? From

$$\hat{k}_t = \sum_{i=0}^{t-1} \left(V_{kk}^i \psi^{\,t-i-1} \right) V_{kz} \varepsilon_1 = \begin{cases} V_{kz} \varepsilon_1 \dfrac{\psi^t - V_{kk}^t}{\psi - V_{kk}}, & \psi \neq V_{kk}, \\ V_{kz} \varepsilon_1 \, t \, \psi^t, & \psi = V_{kk}. \end{cases}$$

if $\psi = V_{kk}$, then

$$V_{kk}^i \psi^{\,t-i-1} = \psi^{\,t-1},$$

so

$$\hat{k}_t = \sum_{i=0}^{t-1} \psi^{\,t-1} V_{kz} \varepsilon_1 = t \psi^{\,t-1} V_{kz} \varepsilon_1.$$

Hence

$$\frac{\hat{k}_{t+1}}{\hat{k}_t} = \frac{(t+1)\psi^t}{t\psi^{t-1}} = \psi \left(1 + \frac{1}{t}\right).$$

For small t, $1 + \frac{1}{t}$ is large, so the ratio exceeds 1 and $\hat{k}_t$ keeps rising.

For large t, $1 + \frac{1}{t} \to 1$, so the ratio tends to $\psi < 1$ and $\hat{k}_t$ falls.

Peak occurs approximately when

$$\psi \left(1 + \frac{1}{t}\right) = 1 \quad \Rightarrow \quad t \approx \frac{\psi}{1 - \psi}.$$

Thus when $\psi = V_{kk}$, $\hat{k}_t$ rises at first, peaks around

$$t \approx \frac{\psi}{1 - \psi},$$

and then decays.

When $\psi \neq V_{kk}$,

$$\frac{\hat{k}_{t+1}}{\hat{k}_t} = \frac{\psi^{t+1} - V_{kk}^{t+1}}{\psi^t - V_{kk}^t} \quad \Rightarrow \quad \frac{\hat{k}_2}{\hat{k}_1} = \psi + V_{kk}.$$

We get an initial rise if $\psi + V_{kk} > 1$.

It peaks when $\hat{k}_{t+1} = \hat{k}_t$, i.e.

$$\psi^{t+1} - V_{kk}^{t+1} = \psi^t - V_{kk}^t,$$

so

$$\psi^t (1 - \psi) = V_{kk}^t (1 - V_{kk}),$$

hence

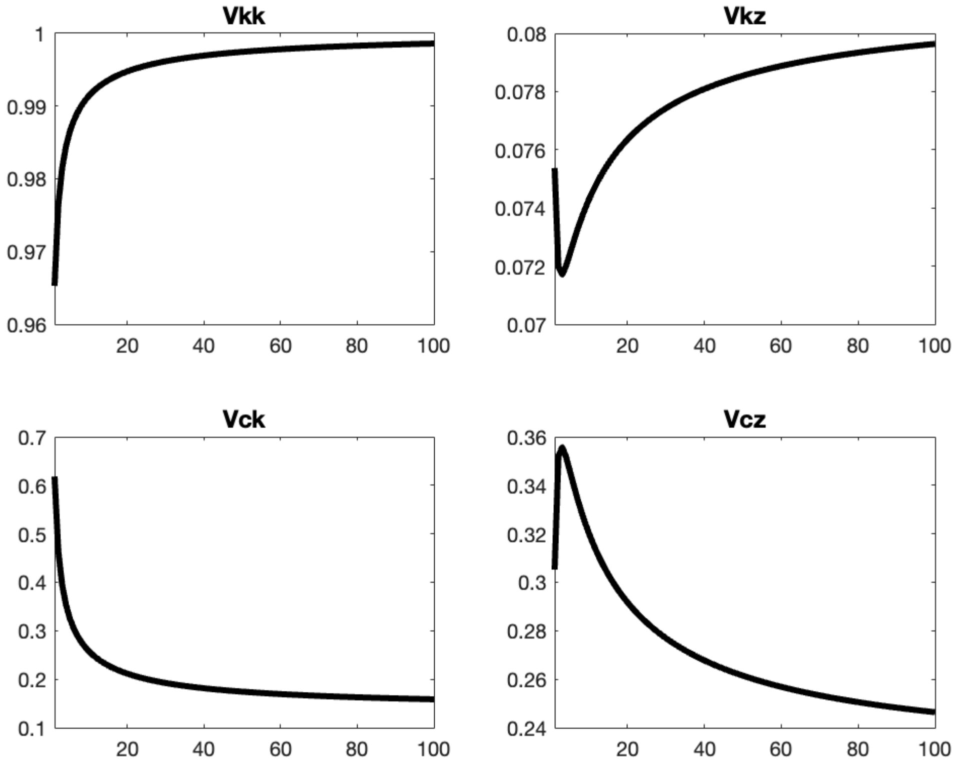
$$t^* = \frac{\ln\left(\frac{1 - V_{kk}}{1 - \psi}\right)}{\ln\left(\frac{\psi}{V_{kk}}\right)}.$$

If $t^* > 1$, the IRF rises for a while and then falls (hump-shaped). If $t^* \leq 1$, it peaks in the first period and then decays monotonically.

The effect of σ

$$\left.\begin{array}{l}\text{Supply only depends on patience } \beta \text{ and } \delta, \text{ labor is fixed.} \\ \text{Demand: investment covers depreciation.}\end{array}\right\} \sigma \text{ does not have long-run effect.}$$

But does have short-run effects, governed by σ.



1.
$$\sigma \uparrow \quad \Rightarrow \quad V_{kk} \uparrow \to 1$$
Capital becomes more persistent and convergence is slower.
Households dislike shifting consumption across time more, i.e. lower IES.

2.
$$\sigma \uparrow \quad \Rightarrow \quad IES = \frac{1}{\sigma} \downarrow \quad \Rightarrow \quad V_{ck} \downarrow .$$
Marginal utility becomes more responsive to changes in consumption, so consumption reacts less for a given state incentive coming from k_{t-1}.

3.
$$\sigma \uparrow \quad \Rightarrow \quad IES \downarrow \quad \Rightarrow \quad V_{cz} \downarrow$$
Consumption's direct response to the shock shrinks.

4.
$$\sigma \uparrow \quad \Rightarrow \quad IES \downarrow \quad \Rightarrow \quad V_{cz} \downarrow \quad \Rightarrow \quad V_{kz} \uparrow$$
After a TFP shock, consumption is more reluctant to jump, so more resources show up as investment / capital accumulation.

Recall the steady state of the system

$$\bar{C} = \bar{Z}\bar{K}^{\alpha} + (1-\delta)\bar{K} - \bar{K},$$
$$1 = \beta\left[\alpha\bar{Z}\bar{K}^{\alpha-1} + (1-\delta)\right].$$

Thus

$$\bar{K} = \left(\frac{\alpha\bar{Z}}{\frac{1}{\beta} - 1 + \delta}\right)^{\frac{1}{1-\alpha}},$$
$$\bar{Y} = \bar{Z}\bar{K}^{\alpha}, \qquad \bar{C} = \bar{Y} - \delta\bar{K},$$

and

$$\frac{\bar{C}}{\bar{K}} = \frac{\bar{Y}}{\bar{K}} - \delta = \bar{Z}\bar{K}^{\alpha-1} - \delta = \frac{\frac{1}{\beta} - 1 + (1-\alpha)\delta}{\alpha}.$$

The effect of δ (depreciation) Example values in the notes:

$$\delta = 0.025: \qquad \bar{K} \approx 38, \quad \bar{Y} \approx 3.7, \quad \bar{C} \approx 2.75,$$
$$\delta = 0.1: \qquad \bar{K} \approx 6.4, \quad \bar{Y} \approx 1.95, \quad \bar{C} \approx 1.3.$$

So higher depreciation implies a lower steady-state level.
Also,

$$\delta \uparrow \Rightarrow V_{kk} \downarrow, \qquad \delta \uparrow \Rightarrow V_{kz} \uparrow.$$

With less capital stock and higher MPK, return and output both respond more proportionally.

For V_{cz}, the sign is ambiguous. The notes list several channels:

- Channel A: $\delta \uparrow \Rightarrow \tilde{\delta} \uparrow$,
- Channel B: $\delta \uparrow \Rightarrow \frac{\tilde{\delta}}{\alpha\beta} - \delta \uparrow$,
- Channel C: $\delta \uparrow \Rightarrow V_{kk} \downarrow$, $V_{ck} \uparrow$.

The final sign depends on calibration.

Complete procedure for solving RBC

1) Model setup: social planner's problem vs. competitive market (isomorphic).
2) Use dynamic programming to derive the necessary conditions(solve the model).
3) Write down the definition of equilibrium, including allocations and equilibrium conditions.
4) Find the steady state.
5) Log-linearize the dynamic system and determine local dynamic property.
6) Use the method of undetermined coefficients to solve the law of motion:

$$k_t = V_{kk}k_{t-1} + V_{kz}z_t,$$

and then solve for the other variables such as c_t.

7) Calibrate and simulate the model.

State Space Approach: 2-D State Space / Saddle Path Analysis

Start from

$$\hat{k}_t = \frac{1}{\beta}\hat{k}_{t-1} - \frac{\bar{C}}{\bar{K}}\hat{c}_t + \frac{\tilde{\delta}}{\alpha\beta}\hat{z}_t.$$

Let $\hat{z}_t = 0$. Then

$$\hat{k}_t = \frac{1}{\beta}\hat{k}_{t-1} - \frac{\bar{C}}{\bar{K}}\hat{c}_t,$$

so

$$\hat{c}_t = \frac{\bar{K}}{\beta\bar{C}}\hat{k}_{t-1} - \frac{\bar{K}}{\bar{C}}\hat{k}_t.$$

Also, from

$$\sigma E_t\hat{c}_{t+1} + \tilde{\delta}(1-\alpha)\hat{k}_t - \tilde{\delta}E_t\hat{z}_{t+1} = \sigma\hat{c}_t,$$

letting $\hat{z}_t = 0$ gives

$$\sigma(\hat{c}_t - \hat{c}_{t+1}) - \big[1-\beta(1-\delta)\big](1-\alpha)\hat{k}_t = 0$$

Rearrange:

$$\begin{cases} \hat{k}_t - \hat{k}_{t-1} = \left(\frac{1}{\beta}-1\right)\hat{k}_{t-1} - \frac{\bar{C}}{\bar{K}}\hat{c}_t, \\ \\ \hat{c}_{t+1} - \hat{c}_t = \frac{1}{\sigma}\big[1-\beta(1-\delta)\big](1-\alpha)\left(\underbrace{\frac{\bar{C}}{\bar{K}}\hat{c}_t - \frac{1}{\beta}\hat{k}_{t+1}}_{=-\hat{k}_t}\right). \end{cases}$$

Steady-state loci If

$$\hat{k}_t = \hat{k}_{t-1},$$

then

$$\hat{c}_t = \left(\frac{1}{\beta}-1\right)\frac{\bar{K}}{\bar{C}}\hat{k}_{t-1}. \tag{1}$$

If

$$\hat{c}_{t+1} = \hat{c}_t,$$

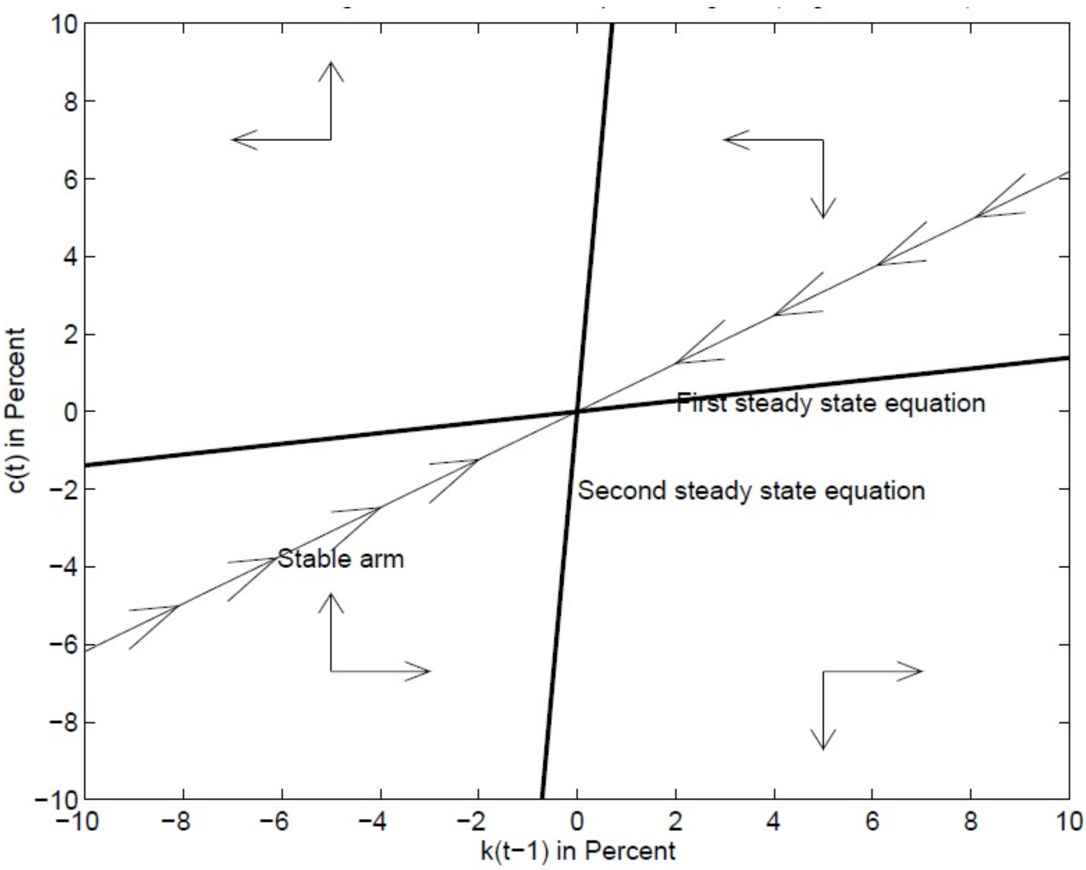
then

$$\hat{c}_t = \frac{\bar{K}}{\beta\bar{C}}\hat{k}_{t-1}. \tag{2}$$

The steady state is

$$\hat{k}_t = \hat{c}_t = 0.$$

The stable arm is the unique saddle path converging to the steady state.



The notes' directional interpretation:

- If $\hat{c}_t$ is above line (1), then $\hat{k}_t < \hat{k}_{t-1}$: consume too much, investment is low, so capital falls.
- If $\hat{c}_t$ is below line (1), then $\hat{k}_t > \hat{k}_{t-1}$: consume too little, save more, so capital rises.
- If $\hat{c}_t$ is above / left of line (2), then $\hat{c}_{t+1} > \hat{c}_t$: consume more next period.
- If $\hat{c}_t$ is below / right of line (2), then $\hat{c}_{t+1} < \hat{c}_t$: consume less next period.

Why These Relations Hold; Redo in Levels

Why line (1) implies $\hat{k}_t < \hat{k}_{t-1}$ If $\hat{c}_t$ is above line (1), then

$$\hat{c}_t > \left(\frac{1}{\beta} - 1\right) \frac{\bar{K}}{\bar{C}} \hat{k}_{t-1},$$

so

$$\frac{\bar{C}}{\bar{K}} \hat{c}_t - \left(\frac{1}{\beta} - 1\right) \hat{k}_{t-1} > 0.$$

Using

$$\hat{k}_t - \hat{k}_{t-1} = \left(\frac{1}{\beta} - 1\right) \hat{k}_{t-1} - \frac{\bar{C}}{\bar{K}} \hat{c}_t,$$

we get

$$\hat{k}_t < \hat{k}_{t-1}.$$

Why line (2) implies $\hat{c}_{t+1} > \hat{c}_t$ If $\hat{c}_t$ is above line (2), then

$$\hat{c}_t > \frac{\bar{K}}{\beta \bar{C}} \hat{k}_{t+1},$$

so

$$\frac{\bar{C}}{\bar{K}} \hat{c}_t - \frac{1}{\beta} \hat{k}_{t+1} > 0.$$

Hence, from

$$\hat{c}_{t+1} - \hat{c}_t = \frac{1}{\sigma} \big[1 - \beta(1-\delta)\big](1-\alpha) \left(\frac{\bar{C}}{\bar{K}} \hat{c}_t - \frac{1}{\beta} \hat{k}_{t+1} \right),$$

we have

$$\hat{c}_{t+1} > \hat{c}_t.$$

Stable arm The stable arm is the unique saddle path converging to steady state:

$$\hat{c}_t = V_{ck} \hat{k}_{t-1}.$$

The speed of convergence is

$$\hat{k}_t = V_{kk} \hat{k}_{t-1} \quad \Rightarrow \quad \hat{k}_{t+h} = V_{kk}^h \hat{k}_t.$$

Nature gives $\hat{k}_{t-1}$, and households choose $\hat{c}_t$ so as to land exactly on the stable arm. Then the economy follows that arm back to steady state. Off the stable arm, the system is unstable and diverges.

Redo in levels Set $Z_t = \bar{Z}$:

$$\begin{aligned} C_t &= \bar{Z} K_{t-1}^{\alpha} + (1-\delta) K_{t-1} - K_t, \\ 1 &= \beta E_t \left(\frac{C_{t+1}}{C_t} \right)^{-\sigma} \left(\alpha \bar{Z} K_t^{\alpha - 1} + 1 - \delta \right). \end{aligned}$$

So

$$K_t - K_{t-1} = \bar{Z} K_{t-1}^{\alpha} - \delta K_{t-1} - C_t,$$

and

$$E_t \frac{C_{t+1}}{C_t} = \left\{ \beta \left[\alpha \bar{Z} \big(\bar{Z} K_{t-1}^{\alpha} + (1-\delta) K_{t-1} - C_t \big)^{\alpha - 1} + 1 - \delta \right] \right\}^{1/\sigma}$$

At steady state, $\Delta K_t = 0$, so

$$C_t = \bar{Z} K_{t-1}^{\alpha} - \delta K_{t-1}. \tag{1}$$

Thus:

$$\begin{aligned} &\text{if } C_t > \bar{Z} K_{t-1}^{\alpha} - \delta K_{t-1}, \quad K_t < K_{t-1}, \\ &\text{if } C_t < \bar{Z} K_{t-1}^{\alpha} - \delta K_{t-1}, \quad K_t > K_{t-1}. \end{aligned}$$

At steady state, $\Delta C_t = 0$, so

$$1 = \left\{ \beta \left[\alpha \bar{Z} \big(\bar{Z} K_{t+1}^{\alpha} + (1-\delta) K_{t+1} - C_t \big)^{\alpha - 1} + 1 - \delta \right] \right\}^{1/\sigma},$$

which implies

$$C_t = \bar{Z} K_{t-1}^{\alpha} + (1-\delta) K_{t-1} - \bar{K}. \tag{2}$$

Then:

$$\begin{aligned} &\text{if } C_t > -\bar{K} + \bar{Z} K_{t-1}^{\alpha} + (1-\delta) K_{t-1}, \quad C_{t+1} > C_t, \\ &\text{if } C_t < -\bar{K} + \bar{Z} K_{t-1}^{\alpha} + (1-\delta) K_{t-1}, \quad C_{t+1} < C_t. \end{aligned}$$

Steady-state equations in levels

$$\begin{cases} C_t = \bar{Z}\, K_{t-1}^{\alpha} - \delta K_{t-1}, & (1) \\ C_t = \bar{Z}\, K_{t-1}^{\alpha} + (1-\delta)K_{t-1} - \bar{K}. & (2) \end{cases}$$

Why is the stable arm still linear? Since

$$\hat{c}_t = V_{ck}\hat{k}_{t-1},$$

we have

$$\frac{C_t - \bar{C}}{\bar{C}} = V_{ck}\frac{K_t - \bar{K}}{\bar{K}}.$$

Thus

$$C_t = \frac{V_{ck}\bar{C}}{\bar{K}}K_t - V_{ck}\bar{C} + \bar{C}.$$

So in levels the stable arm is linear, with slope

$$\frac{V_{ck}\bar{C}}{\bar{K}}.$$